Rotaract Club of Xiamen University Malaysia

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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Hokkien Mee Is Purple
Project / Solution Name	Varity
Name (Leader)	Tay Ee Zhan
Email & Phone (Leader)	eezhan0805@gmail.com / +60 11-6561 6811
University Affiliation(s)	Universiti Malaya (both members)
Live Prototype URL	https://verity--eezhan0805.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/TayEeZhan/varity (public; Apache License 2.0)

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

User (browser) → Varity web app (static HTML/CSS/JS) → FastAPI server on a Replit Autoscale deployment → engine modules (Python) → committed data files in data/.
Data flow: the engine downloads every Hacker News "Who is hiring?" thread via the Algolia HN Search API (94 threads, 48,508 posts, Jan 2019 – Oct 2026). A tagger finds 40 taxonomy skills per post; the engine builds a monthly share per skill, forecasts 12 months ahead with a conformal band and a Welch trend label, and joins AI exposure from the Anthropic Economic Index. The role module learns role profiles; the pathway module orders the learning plan.
CV upload → Claude Haiku 5.5 (Anthropic API) points at skills with quotes → code checks each quote word for word → role match.
Accounts: optional Replit Auth sign-in; profiles in Replit PostgreSQL (SQLite fallback). Secrets in Replit Secrets.
The repository holds only aggregates; raw post text is never committed.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12	Language	PSF License
FastAPI ≥0.115 + uvicorn ≥0.30	Web server and JSON APIs	MIT / BSD-3-Clause
pandas 3.0.6, NumPy 2.5.4	Tables and forecast maths	BSD-3-Clause
SciPy 1.18.1, statsmodels 0.15.0	Welch t-test; damped-trend model (backtest only)	BSD-3-Clause
anthropic ≥2.0.0, pydantic ≥2.10	Claude API client; validates model output	MIT

Name & Version	Purpose in Project	License
pypdf ≥6.21	Reads uploaded PDF CVs	BSD-3-Clause
requests 2.34.2, httpx ≥0.28	HTTP calls (Algolia HN API, feeds)	Apache-2.0 / BSD-3-Clause
Jinja2, APScheduler, feedparser, PyYAML, Markdown	Signal-ranker pages, schedule, RSS, config	BSD / MIT
replit-object-storage 1.0.2	Replit App Storage client	ISC
scikit-learn 1.9.1	Gradient-boosting experiment (not shipped)	BSD-3-Clause
pytest 9.1.1	700+ automated tests	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Anthropic Messages API (Claude Haiku 5.5)	CV skill intake, CV generation, optional route advisor and pathway coach ($1/day cap)	Paid (team member's own account)
Jev (TypeSafe AI) via Replit AI Integrations	Evaluated as a news ranker; shown as a second opinion on Signals	Sponsor-provided (Replit)
OpenRouter via Replit AI Integrations	3-model AI jury that graded the Signals ranker (evaluation only)	Sponsor-provided (Replit)
Algolia HN Search API	Fetches Hacker News 'Who is hiring?' threads	Free, public, no key
Replit Auth (OpenID Connect)	Optional user sign-in	Sponsor-provided (Replit)
Google Tasks API + Google Identity Services	Optional 'Send today to Google Tasks' (runs in the user's browser)	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Autoscale Deployment	Hosts the FastAPI server and web app at https://verity--eezhan0805.replit.app	SDC-provided Replit credits (US$20) / Asia
Replit PostgreSQL	Accounts and saved profiles	Included with Replit plan
Replit Secrets	Holds ANTHROPIC_API_KEY, SESSION_SECRET, DATABASE_URL	Included
Replit App Storage + Scheduled Deployment	Not used (optional daily-refresh code exists but is not set up)	None

Service & Provider	Purpose in Project	Plan / Region

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Claude Haiku 5.5 (claude-haiku-5-5)	Anthropic API	Pre-trained	Anthropic commercial terms
Naive-3 forecast + conformal band + Welch t-test	Our code (SciPy)	Self-built statistical model	Apache-2.0 (ours)
Learned role-skill profiles (9 roles)	Our code, from HN posts	Self-trained	Apache-2.0 (ours)
HistGradientBoostingRegressor (scikit-learn)	scikit-learn	Self-trained; experiment only, not deployed	BSD-3-Clause
Jev decision model (jev-1.13.0)	TypeSafe AI, via Replit AI Integrations	Pre-trained; tested as ranker (NDCG@10 0.25 vs formula 0.53), shown as second opinion	TypeSafe API terms
AI jury: Claude Sonnet 5.5, GPT-6.1 Sol, Gemini 4 Argon	OpenRouter via Replit AI Integrations	Pre-trained; evaluation only (labels stories for NDCG@10)	Each provider's API terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Hacker News 'Who is hiring?' threads	https://hn.algolia.com/api	Public	None stated; aggregates only committed	Y in raw text (never committed); N in committed files
Anthropic Economic Index 2026-06-26	https://huggingface.co/datasets/Anthropic/EconomicIndex	Public	CC BY / MIT	N
O*NET 31.0 + O*NET-SOC crosswalk	https://www.onetcenter.org/database.html	Public	CC BY 4.0	N
GPTs are GPTs; AIOE; Microsoft Working with AI	github.com/openai/GPTs-are-GPTs; github.com/AIOE-Data/AIOE; github.com/microsoft/working-with-ai	Public (cross-check only)	MIT; none (cited); CC BY 4.0	N
Example profiles, synthetic CVs, 77 learning links	data/example_profiles.json, demo-cvs/, data/resources.json	Synthetic / self-collected	Ours	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Claude Opus 5.5, Sonnet 5.5, Haiku 5.5)	Code, tests, data pipeline and documentation from the team's prompts and plans	Major
Replit Agent	Replit environment setup (.replit file only)	Minor
ChatGPT (OpenAI)	Asking general questions	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

A second model reviews each major part, and the team reviews every pull request. We run over 700 automated tests, including tests that block look-ahead leaks. Every statistic traces to a file in data/ or a cited source. Reviews found real faults that we fixed: bare "AI" inflated Machine Learning to 55% of posts, and the first uncertainty band covered only 65.9% of real values. Both members can explain each part.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Platform (deployment, Auth, PostgreSQL, Secrets)	Hosts the live prototype, optional sign-in and account storage
Replit	AI Integrations (Jev; OpenRouter models)	Ran Jev (second-opinion ranker) and the 3-model AI jury (Claude Sonnet 5.5, GPT-6.1 Sol, Gemini 4 Argon) that evaluated the Signals ranker

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit hosts the live prototype, sign-in and account database, so it is essential to the deployed demo as built. It is replaceable: the server is plain FastAPI, and engine/store.py has a local-folder store and SQLite fallback, so any Python host would work. RunPod is not used because Varity does no GPU work.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Sign-in is optional. For a signed-in user we store the Replit user id, email, display name and profile image link (personal data under Malaysia's PDPA 2010) in Replit PostgreSQL, to keep their profile and progress. The user can delete the account at any time; we sell or share nothing. A profile holds the skill list, target role and ticks. For CV generation we also keep the cleaned CV text (emails, phone and ID numbers already redacted), deleted with the profile. CV text is sent to Anthropic; redaction happens first. The demo uses synthetic CVs only. Raw Hacker News posts are never committed; a test checks committed data holds no email addresses.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

1. Hacker News skews US and startup, not Malaysian; the app says so. 2. Simple persistence beat every trend model at 12 months; we ship it with a tested band rather than over-claim. 3. AI-exposure indices disagree; we show them as estimates. 4. The LLM can miss or invent a skill: code drops any skill whose quote is not in the CV, and the user edits the list. 5. Single LLM provider: with no key, a $1/day cap, or an error, code takes over (keyword pass, formula ranking). 6. No Malaysian job feed: JobStreet blocks bots and we bypass nothing.

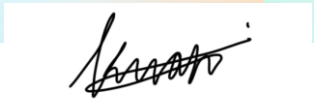
7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature:  Tay Ee Zhan (Team Leader)  Goh Sheng Kuan (Member)

Date: 25 October 2026